

Generative Interpretability via Scalable Neuro-Symbolic Models

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Post-hoc interpretability cannot audit

- Explain but do not control. Internal states locate a model's own errors near-perfectly, yet steering on them fails to correct behaviour.
- Do not survive deployment. SAE and cross-layer-transcoder dictionaries run to millions of features and are discarded before release.
- Arrive too late. An explanation exists only once the unsafe output does; CoT steps are tokens from completed passes, and unfaithful to the computation.

DARPA–NSF–CAISI AI Forge names interpretability one of three priority areas; its thrust is titled “Enabling Actionable Understanding”.

Generative interpretability

An architectural property under which a model's inference pass exposes intermediate checkpoints that are (i) semantically meaningful to humans natively and (ii) causally intervenable by construction.

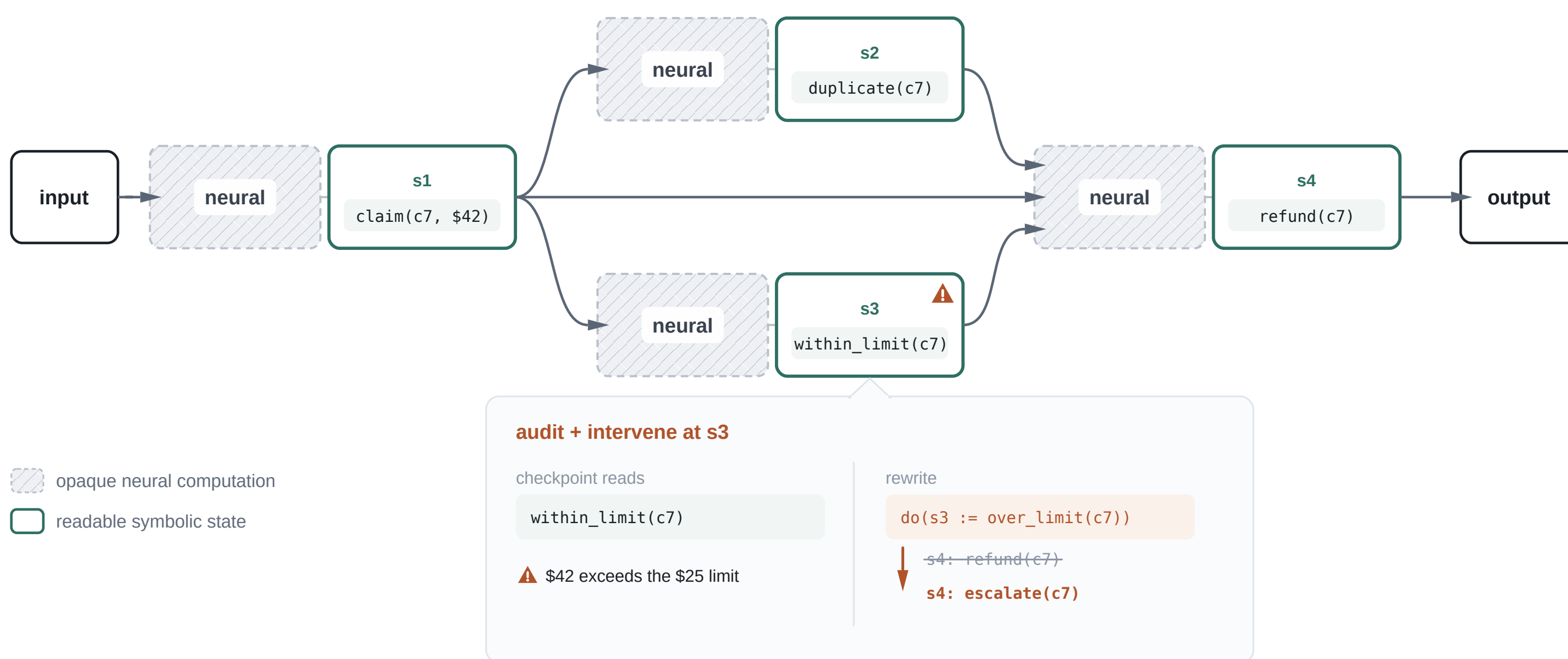
- Native meaning: a checkpoint maps to a human-readable concept with no external tool to recover it — mathematical transparency alone does not qualify.
- Causal intervention: checkpoints are variables the pass functionally depends on, so $\text{do}(\cdot)$ yields well-defined counterfactuals rather than approximate effects.

Two instances with generative interpretability

Both preserve the computational topology that circuit tracing finds in LLMs, substituting native checkpoints for a subset of its feature nodes.

Neuro-Symbolic Model

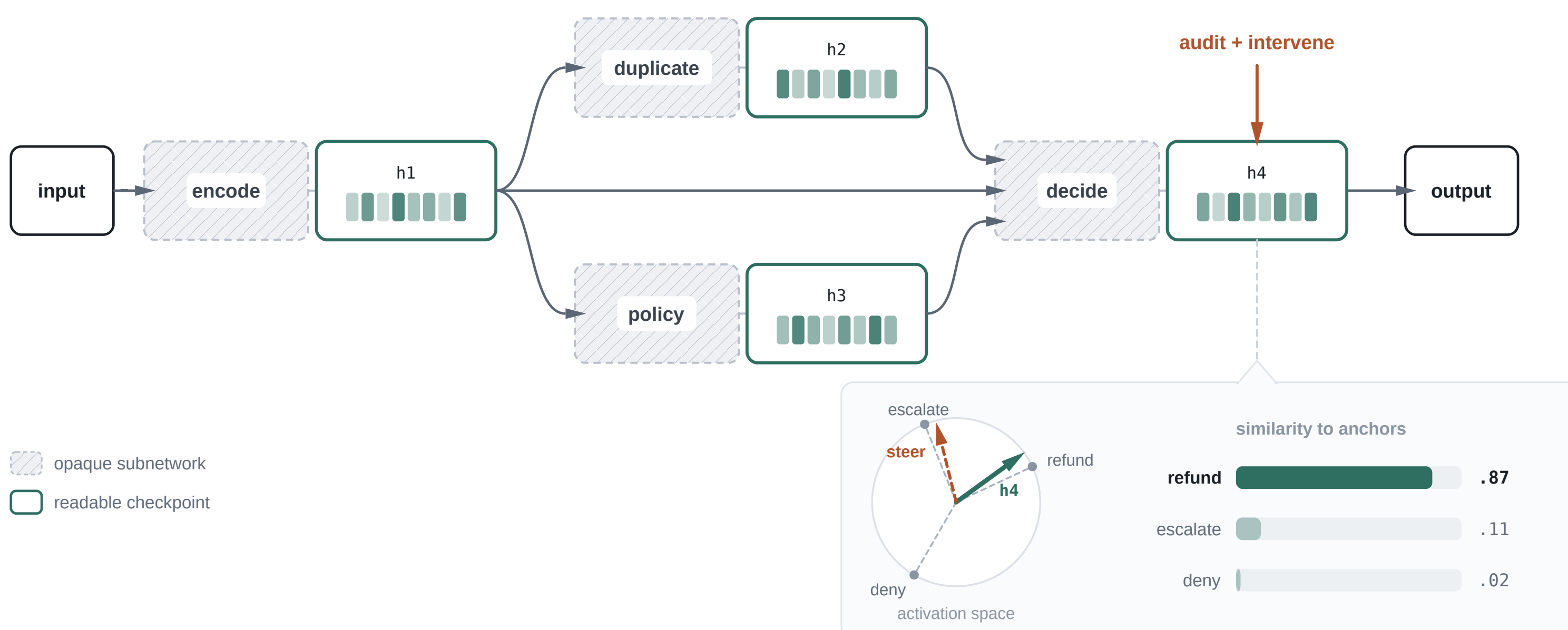
a DAG of steps — neural and symbolic components alternate, every symbolic state is readable



Symbolic states as checkpoints: each is a proposition; the audit checks entailment, and the rewrite re-fires the rule downstream.

Modular Neural Network

a graph of specialised subnetworks — outputs stay continuous vectors, read by similarity to anchors fixed by each module's training objective



Subnetwork outputs as checkpoints: vectors read by similarity to the module's anchors; the audit steers the decision to escalate.

What generative interpretability buys for agent safety

Attacks must become legible

Indirect prompt injection can no longer act on an opaque policy: it has to render itself in a schema a monitor inspects before anything commits.

Responsibility becomes attributable

An undesired outcome traces to the checkpoint that bears it — and, for modules, to whoever trained it, making model supply chains auditable.

Audit becomes externally checkable

Checkpoints are natively legible rather than reconstructed by whoever holds the tools, so an audit yields an artifact third parties can verify.



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